Supplementary Materials

Individual differences in the computational processes that support face recognition

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Priors

These are the priors used for the preregistered LBA model.

parameter	description	condition	correct	response	mean	sd
v_SNovel:lMFALSE	Drift rate	Novel	False	-	0	2
v_SLearned:lMFALSE	Drift rate	Learned	False	-	0	2
v_SNovel:lMTRUE	Drift rate	Novel	True	-	1	2
v_SLearned:lMTRUE	Drift rate	Learned	True	-	1	2
sv_lMTRUE	SD drift rate (true accumulator)	-	True	-	log(0.5)	0.5
B_lRNovel	Threshold	-	-	Novel	log(1)	0.5
B_lRLearned	Threshold	-	-	Learned	log(1)	0.5
A	Start point	-	-	-	log(0.5)	0.3
t0	Non-decision time	-	-	-	log(0.3)	0.3

Model checks

Convergence

The chains showed satisfactory mixing as evidenced by the rhat values.

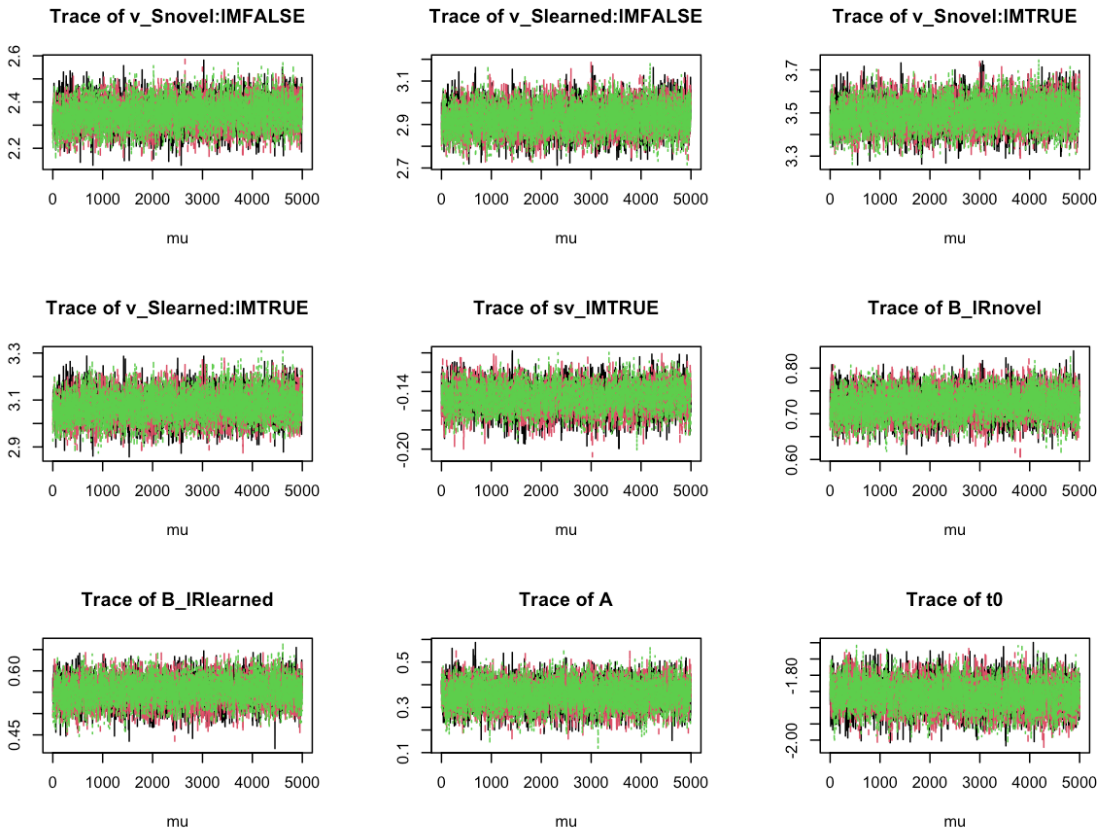
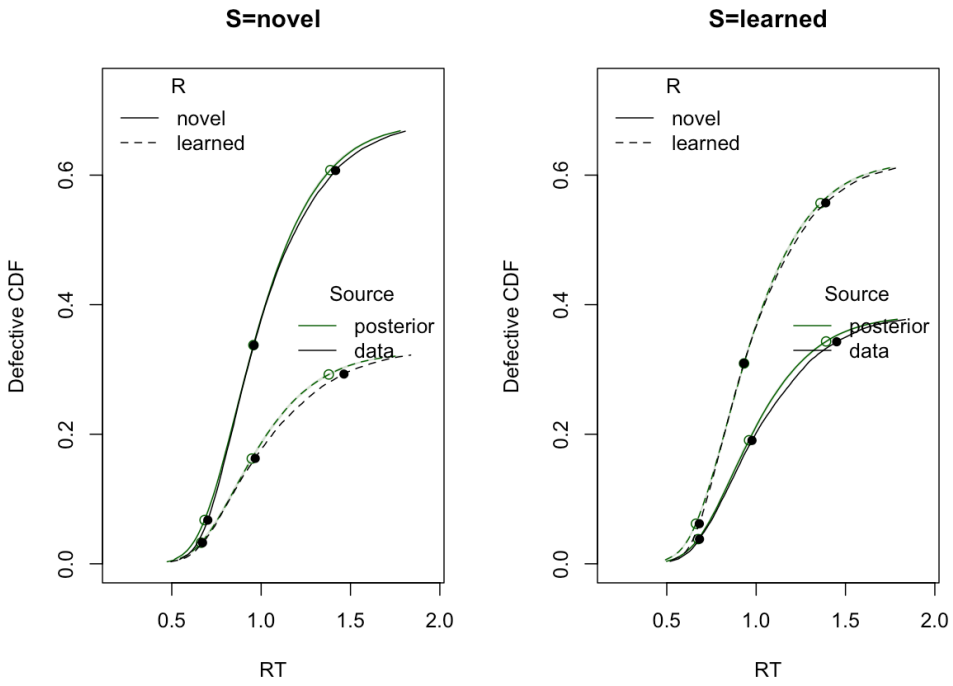
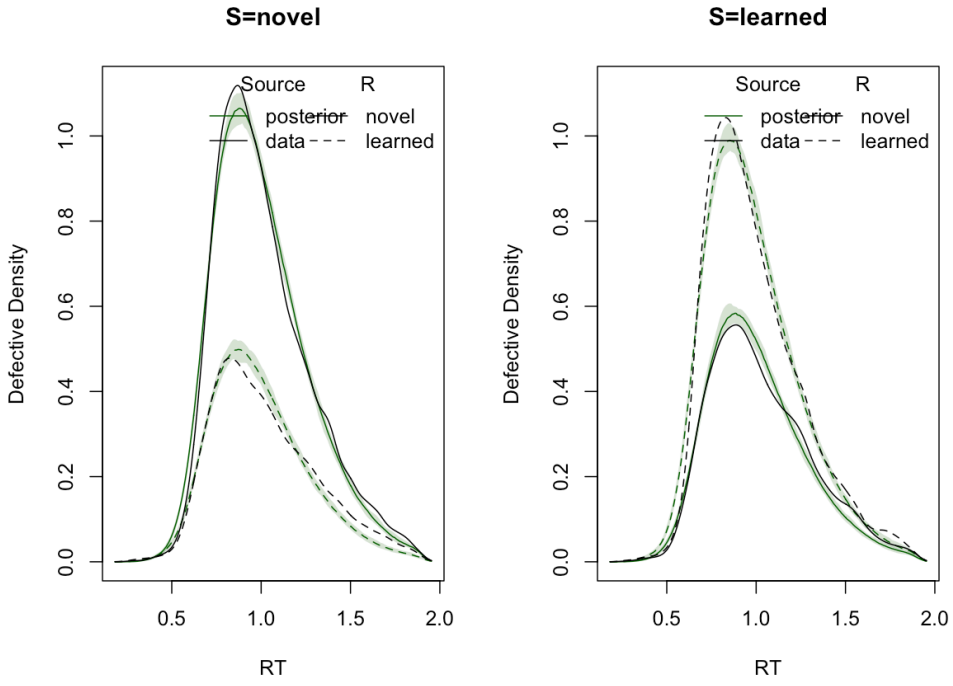


Figure 1. MCMC chains

Posterior predictive checks



(a) CDF



(b) Density

Figure 2. Posterior predictive checks

Prior vs posterior plots

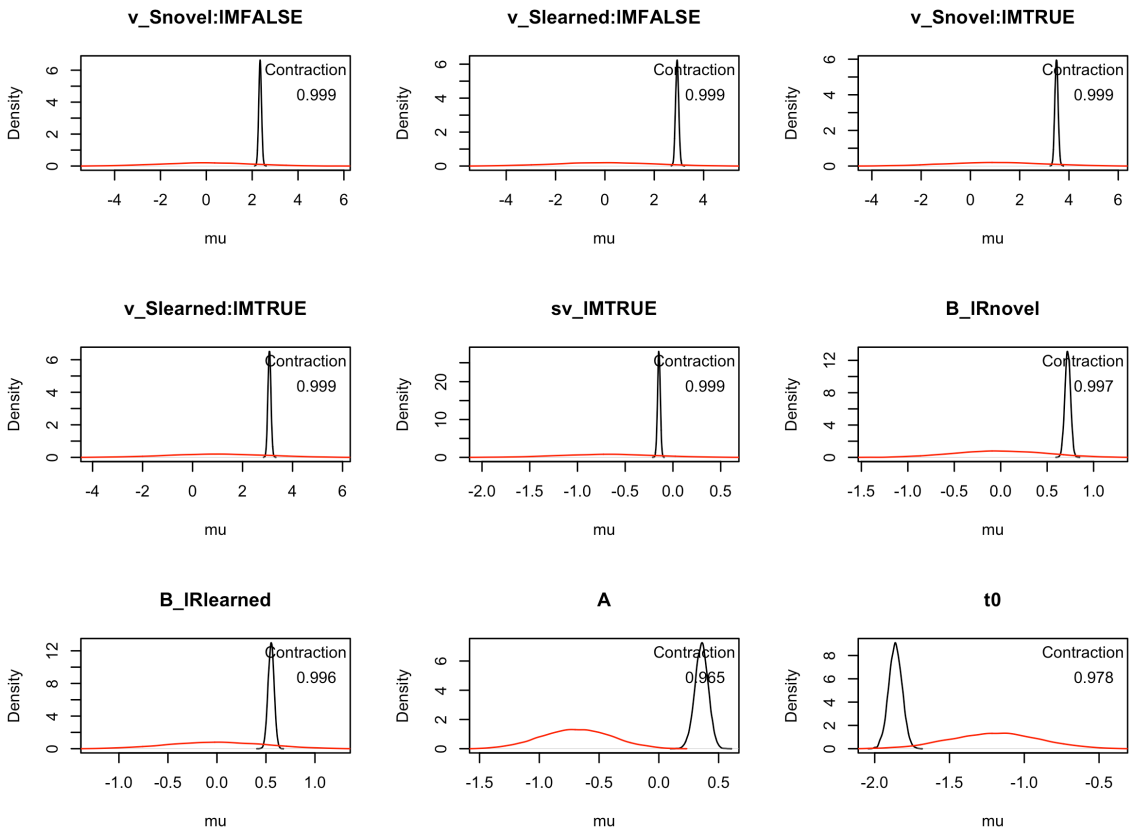


Figure 3. Priors vs posterior plots

Subject-level drift rates and thresholds

To visualize the individual variability of our parameters of interest, distributions of the means of the subject-level posterior distributions for threshold (Learned and Novel) and drift rate (Learned and Novel) can be found in Figure 4.

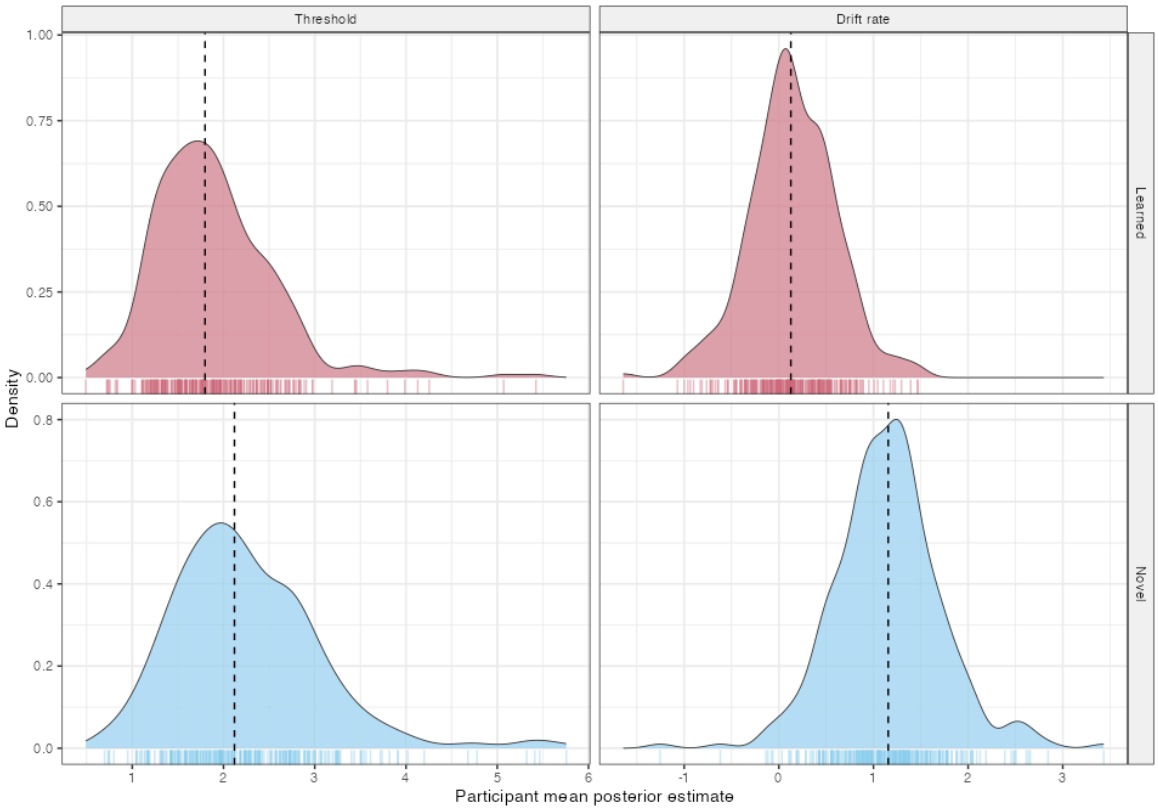


Figure 4. Distribution of subject-level posterior means of the threshold parameter (Learned and Novel) and the drift rate parameter (Learned and Novel). Black doted lines denotate the median parameter estimates and the small color lines denotate the individual subject mean parameter estimates.

To assess whether participants' parameter estimates varied enough to support an individual-differences approach, we compared the spread of the subject-level posterior means with the precision of each individual estimate. Estimates from hierarchical models are shrunk toward the group mean, so their observed spread understates true between-participant variance. More importantly, variability is only meaningful when it exceeds the uncertainty of each estimate (Haines et al., 2025; Hedge et al., 2018; Rouder & Haaf, 2019). For each parameter and condition, we therefore computed the ratio of the between-participant SD of posterior means to the average within-participant posterior SD. Ratios were 1.88-2.04 for threshold and 1.307–1.535 for drift rate, indicating that participants differed by more than the uncertainty in their individual estimates.Together, these results indicate substantial between-participant variability.

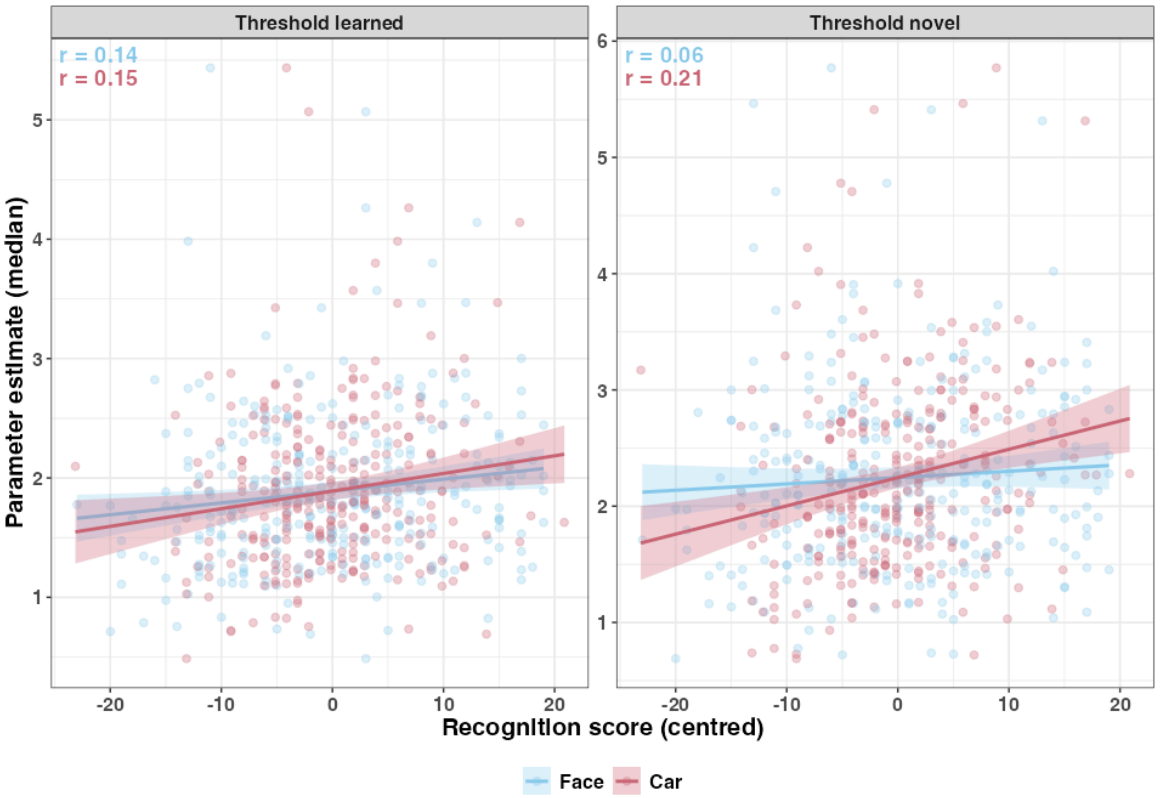
Table 1. Between-Participant Variability in Subject-Level LBA Parameter Estimates Relative to Within-Participant Estimation Uncertainty

		Variability vs. precision		
Parameter	**Condition**	**Between**	**Within**	**Ratio**
Threshold	Learned	0.659	0.351	1.879
Threshold	Novel	0.800	0.391	2.045
Drift rate	Learned	0.457	0.349	1.307
Drift rate	Novel	0.565	0.368	1.535

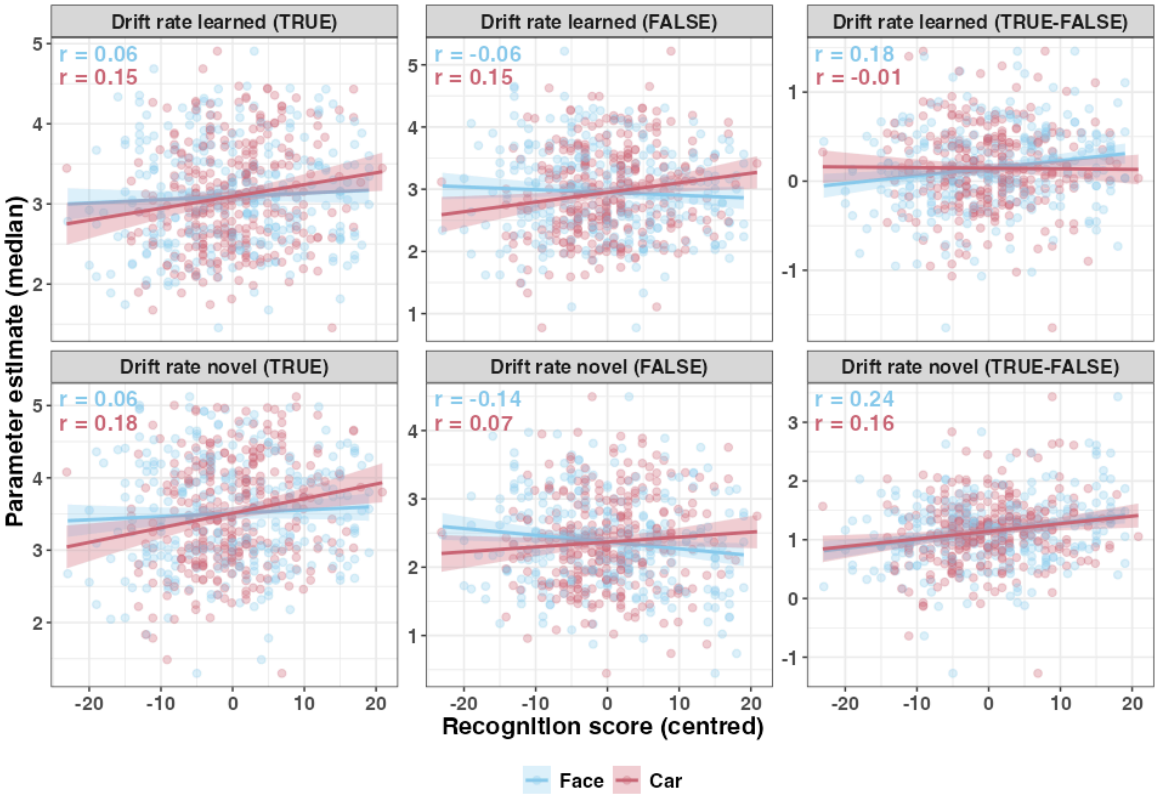
Between = standard deviation of participants' posterior means; Within = average within-participant posterior standard deviation; Ratio = Between / Within.

Point estimate correlations between LBA parameters and Cambridge Car Memory score/Cambridge Face Memory scores

To estimate the association between LBA parameters and car recognition ability, we computed the linear relationship between the median of the posterior distribution of each estimated parameters of interest and the car recognition ability scores centred (z-transformed). These relationships can be seen in Figure 5. Correlations with the face recognition scores were added to facilitate comparisons between these correlations



(a) Median threshold correlations with recognition ability



(b) Median drift rate correlations with recognition ability

Figure 5. Point estimates correlations with recognition scores. In (b), drift rate (TRUE) reflects the accumulation of evidence when the response is correct and drift rate (FALSE) the accumulation of evidence when the response is incorrect. Drift rate (TRUE-FALSE) represent the signal-to-noise ratio provided by the stimulus.

Plausible value correlations with Cambridge Car Memory scores and Cambridge Face Memory scores

We calculated the posterior correlations distributions between car recognition ability and our parameters of interest. Correlations with threshold can be found in Figure 6 and correlations with drift rate in Figure 7. Table 2 summarize these distributions. The posterior correlations with face recognition ability were added to facilitate comparisons between the two memory scores.

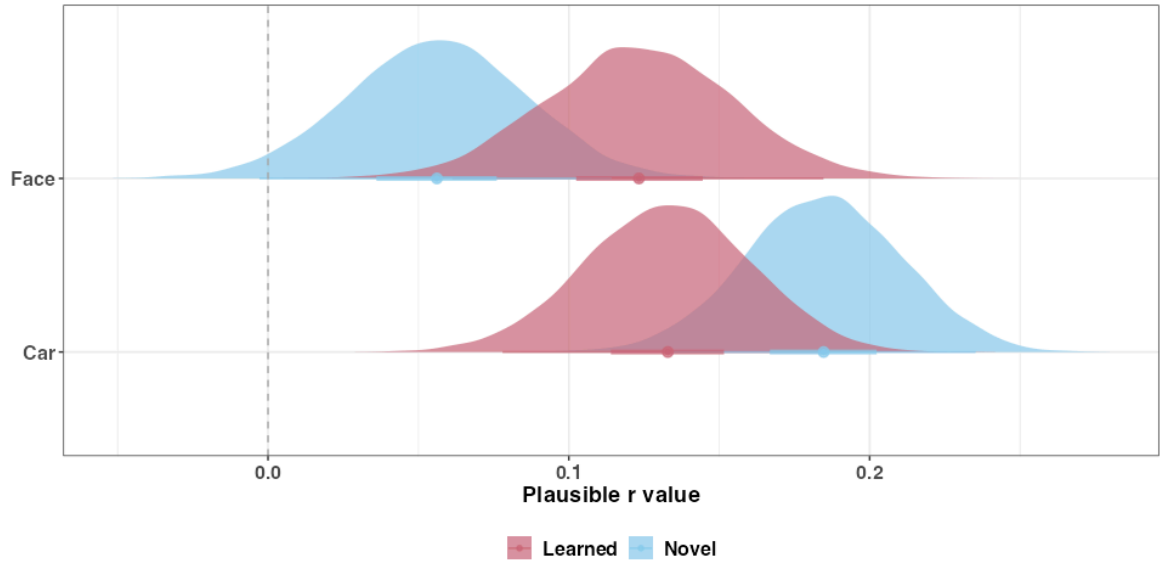
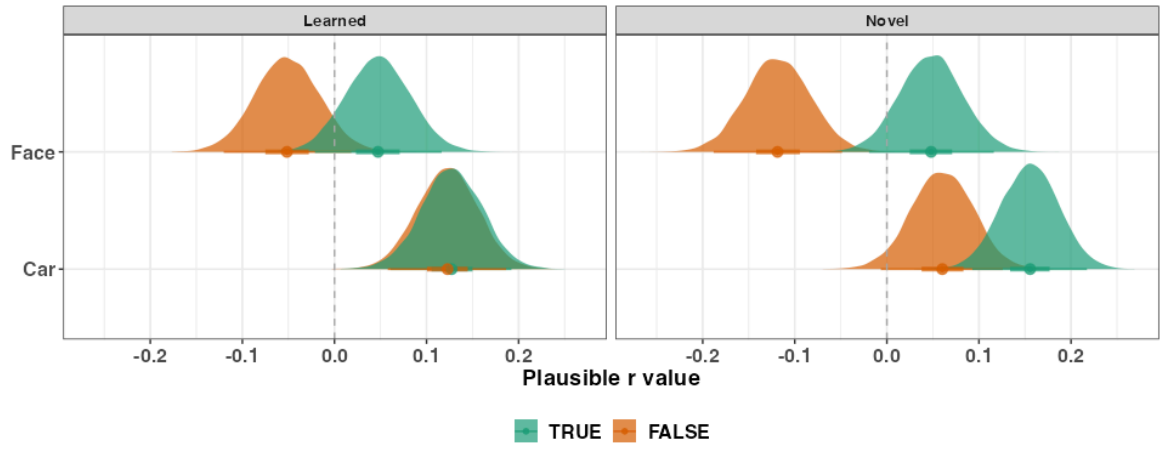
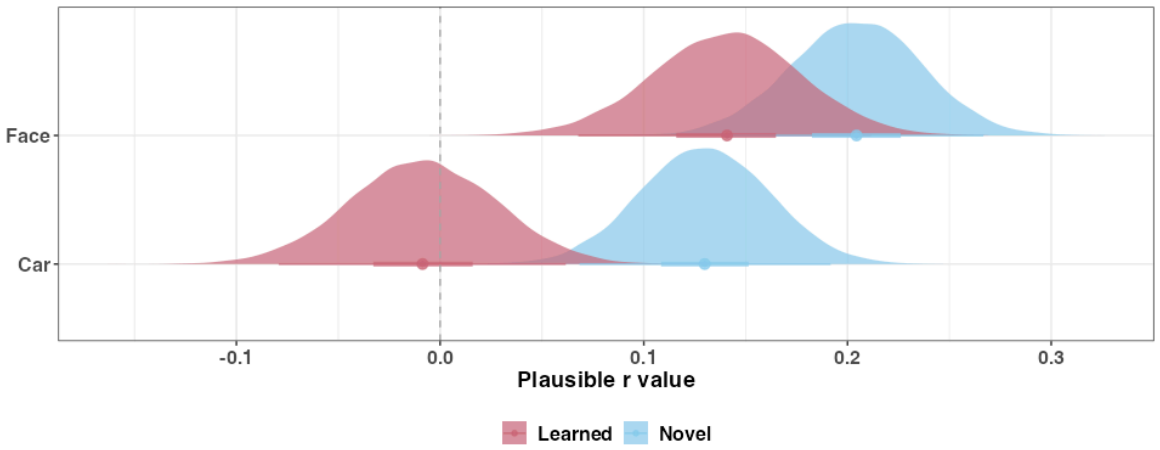


Figure 6. Plausible correlation values distributions between threshold by response and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.



(a) Plausible value correlations between drift rate (TRUE) and drift rate (FALSE) and recognition abilities for both stimulus conditions



(b) Plausible value correlations between drift rate (TRUE-FASLE) and recognition abilities for both stimulus conditions

Figure 7. Plausible correlation values distributions between drift rate parameters by stimulus and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

Table 2. Median r and 95% quantile intervals of the plausible value correlations between LBA parameters and recogntion ability scores.

			Face		Car	
Parameter	**Condition**	**Accuracy**	**median r_Face**	**95% QI_Face**	**median r_Car**	**95% QI_Car**
Threshold	Learned	-	0.12	[0.06, 0.18]	0.13	[0.08, 0.19]
Threshold	Novel	-	0.06	[−0.00, 0.11]	0.18	[0.13, 0.24]
Drift	Learned	TRUE	0.05	[−0.02, 0.12]	0.13	[0.06, 0.19]
Drift	Learned	FALSE	−0.05	[−0.12, 0.02]	0.12	[0.06, 0.19]
Drift	Learned	TRUE-FALSE	0.14	[0.07, 0.21]	−0.01	[−0.08, 0.06]
Drift	Novel	TRUE	0.05	[−0.02, 0.12]	0.16	[0.09, 0.22]
Drift	Novel	FALSE	−0.12	[−0.19, −0.05]	0.06	[−0.01, 0.13]
Drift	Novel	TRUE-FALSE	0.20	[0.14, 0.27]	0.13	[0.07, 0.19]

Note. Blue = 95% QI entirely above zero; red = 95% QI entirely below zero.

Bayesian linear regression models for threshold parameter

We ran two Bayesian linear regression models to estimate if face recognition ability could predict the threshold parameter (response cautions) for both the Learned and Novel responses. The models description can be found in Table 3.

Following our preregistration, the predictions of the two regression models for threshold can be seen in Figure 8. From visual inspection, the response caution seems to increase when face recognition increases for both the Learned and Novel response. This pattern of prediction remains the same when accounting for car recognition ability for the “Learned” response but not for the “Novel” response. Inspection of the posterior predictions in Table 4 shows that some of these predicted intervals overlap with 0, and that caution is needed when extrapolating these results.

Table 3. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median threshold	response	Face	Median threshold: 0 + response + face_ability + (1|participant)
Median drift rate	response	Face + Car	Median drift rate: 0 + response + face_ability + car_ability + (1|participant)

response = two-choices response (Learned vs. Novel)

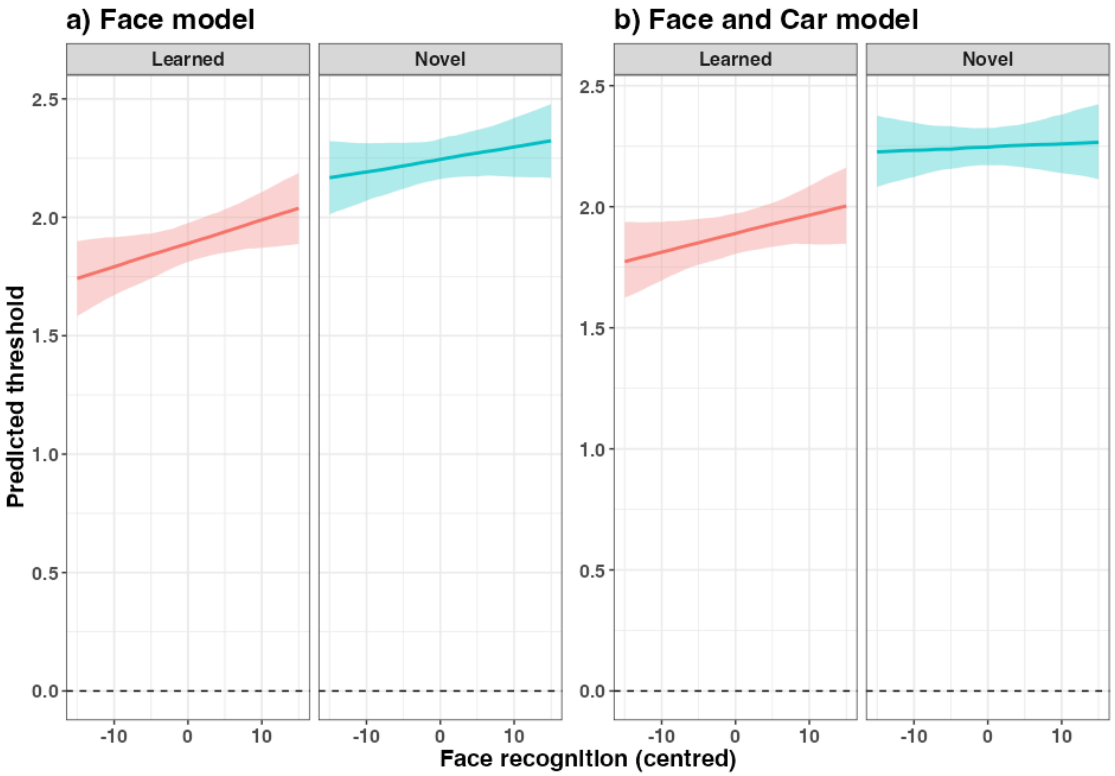


Figure 8. Predicted threshold values from face recognition ability scores for a) Face model and b) Face and Car model

Table 4. Estimates and 95% quantile intervals of the predicted threshold values slope from face recognition ability scores

Parameter	Model	Response	Estimate	l-95% CI	u-95% CI
Threshold slope	Face	Learned	0.0098	0.0009	0.0185
Threshold slope	Face	Novel	0.0053	−0.0035	0.0136
Threshold slope	Face + Car	Learned	0.0077	−0.0012	0.0161
Threshold slope	Face + Car	Novel	0.0012	−0.0076	0.0106

Bayesian linear regression models: Car predictor

As an exploratory analysis, we were interested to see how does object recognition ability predict the drift rate and threshold values estimated from our face recognition task. We ran two Bayesian linear regression models: the first one to estimate if object recognition ability could predict the response caution for learned and novel response and the second one to estimate if object recognition ability could predict the different drift rates (TRUE, FALSE and TRUE-FALSE) for learned and novel face stimuli. The models specificity can be found in Table 5.

Table 5. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median threshold	response	Car	Median threshold: 0 + response + car_ability + (1|participant)
Median drift rate	familiarity:accuracy	Car	Median drift rate: 0 + familiarity:accuracy + car_ability + (1|participant)

response = two-choices response (Learned vs. Novel); familiarity = face stimulus familiarity (Learned vs. Novel); accuracy = response accuracy (TRUE vs. FALSE)

Threshold: Regarding the threshold (response caution) predictions, visual inspections of the predictions show that as car recognition ability increases, the response caution for both, the Learned and Novel responses, increases (Figure 9).

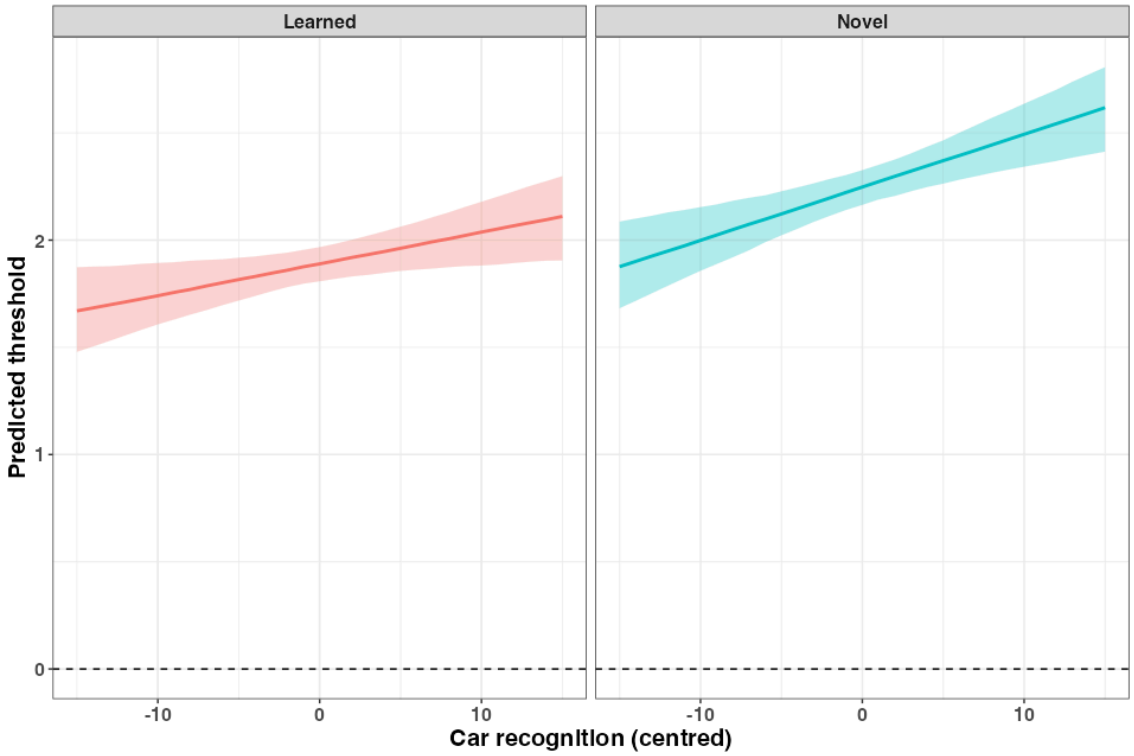


Figure 9. Predicted threshold values from car recognition ability scores for learned and novel responses

Drift rate (TRUE and FALSE): Regarding the drift rate (TRUE) and (FALSE), visual inspections of the predictions show that, as car recognition ability increases, both drift rate (TRUE and FALSE) for both stimuli conditions (Learned and Novel) increases as well. In summary, as car recognition ability increases, the accumulation of signal (drift rate TRUE) and the accumulation of noise (FALSE) increase, regardless of the stimulus condition (Figure 10).

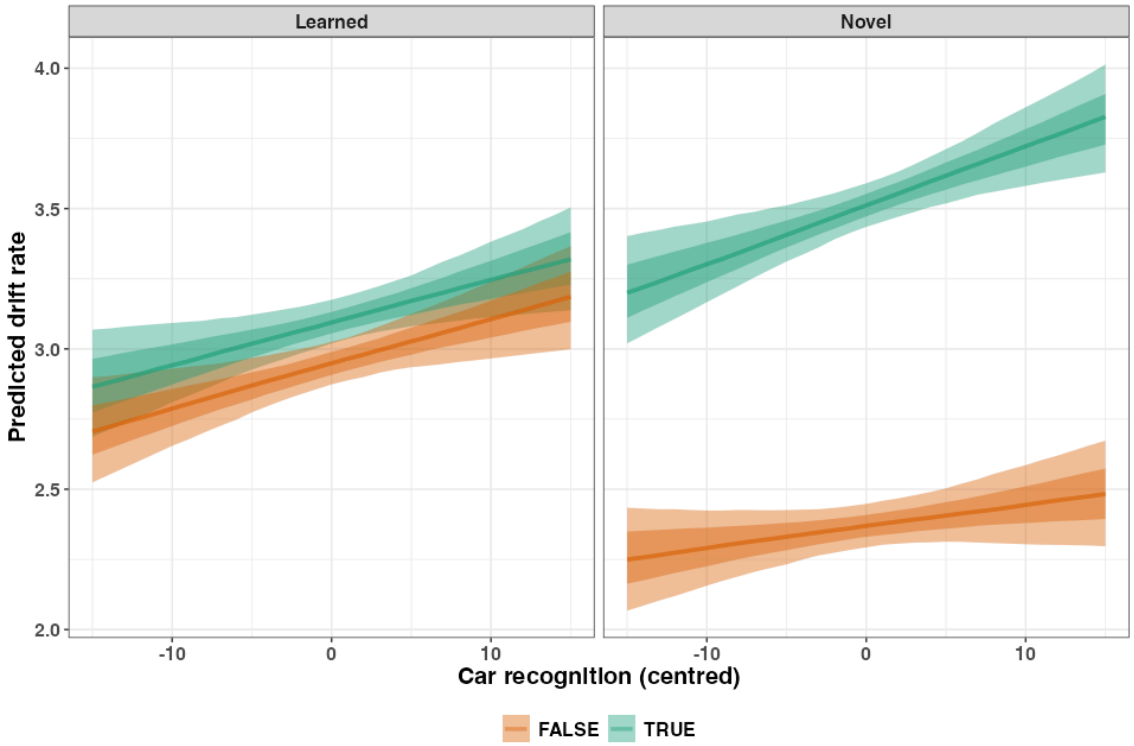


Figure 10. Predicted drfit rate (TRUE and FALSE) values from car recognition ability scores for learned and novel faces

Drift rate (TRUE-FALSE): Regarding the drift rate (TRUE-FALSE), or the signal-to-noise ratio provided by the stimulus, visual inspections of the predictions show that, as car recognition ability increases, only the drift rate for Novel faces increases, whereas the drift rate for Learned face does not follow this pattern. In summary, as car recognition ability increases, the ability to discriminate Novel faces increases but not the ability to discriminate Learned faces (Figure 11).

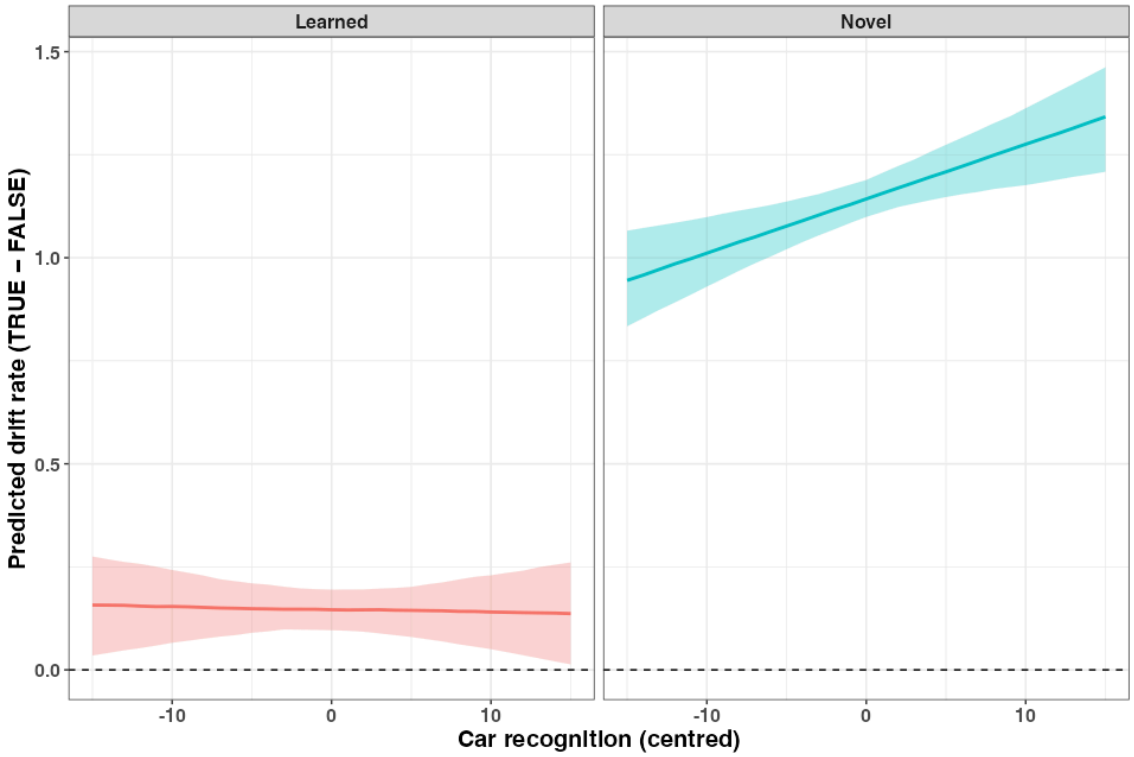


Figure 11. Predicted drift rate (TRUE-FALSE) values from car recognition ability for learned and novel faces

References

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